Preference Learning versus Congestion:

Welfare Effects of Sequential Admissions Mechanism in France

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Abstract

Most centralized admission systems require applicants to commit to a ranking before they know which programs they prefer or can attain. An alternative design lets applicants apply before committing to any rank-ordered list, observe which programs are attainable, and learn their preferences over the course of the process. But clearing in real time within a bounded window can generate congestion that falls disproportionately on lower-ranked candidates. I quantify this tradeoff using administrative records from *Parcoursup*, France's centralized admission platform for post-secondary education, that track each applicant's full decision path—every application, offer, and response. I recover valuations for programs at two stages: at application, from revealed-preference inequalities on portfolio choice under admission risk; at enrollment, from the observed hold-and-accept sequence. A counterfactual decomposition isolates a flexibility value—the welfare gain from responding with informed rather than uninformed preferences—and a congestion cost—the welfare loss from truncating the sequential clearing process. Preliminary estimates indicate that the taste shocks realized during the response window are several times as dispersed as those known at application, evidence that much of the preference relevant to enrollment is learned only after applications are submitted.

1 Introduction

In most countries, it has become increasingly common to use a centralized clearinghouse to assign students to schools. The dominant design is the following: each applicant submits a rank-order list (ROL) of programs, and an algorithm computes an assignment based on each program's evaluation of its applicants. The implicit assumption for the mechanism described above to work well is that applicants know their own preferences over the options and understand the mechanism in place. A growing body of evidence suggests, however, that this premise often fails. Applicants struggle both with holding correct beliefs about mechanisms that require strategic play (Kapor et al., 2020), and with learning about their own preferences over options that have not yet become concrete (Grenet et al., 2022; Hakimov et al., 2023). If preferences only crystallize once options become more or less attainable, then a mechanism that forces students to rank in the dark may produce welfare losses—even if the algorithm itself has desirable theoretical properties.

A different class of mechanisms turns this rigidity into an advantage, letting participants learn their preferences over the course of the procedure and act on what they learn. The leading example is *decentralized deferred acceptance* (sequential DA hereafter), studied by Roth and Xing (1997), in which the match unfolds in real time: applicants send their files to a dozen sites, institutions evaluate the files and make offers down their rankings, applicants respond as offers arrive, and the assignment emerges from these moves. In principle, both static and sequential DA implement the college-optimal stable matching—but only when there is **enough time** for the market to clear. In practice, that condition can fail. Simulating the preferences of participants in the clinical-psychology market, Roth and Xing (1997) find that the procedure takes on average over 18 hours to reach a stable outcome when each offer takes five minutes to process, far longer than the 7 hours the market is open. Sequential clearing that runs out of time before it converges leaves participants with worse assignments than the algorithm would otherwise deliver, a shortfall also documented in Abdulkadiroğlu et al. (2017).

The two designs thus trade off against each other. The static mechanism clears quickly but forces students to commit before they know what they want; the sequential mechanism lets preferences form as offers arrive but pays for that flexibility in congestion. This paper weighs the two forces against each other: whether the preference learning a sequential mechanism affords is worth the congestion it incurs, and how the gains and losses are distributed across participants.

I study this tradeoff using *Parcoursup*, the platform through which essentially all of France's roughly 600,000 annual high school graduates are assigned. *Parcoursup* is a sequential implementation of college-proposing deferred acceptance, similar to the psychologist market studied in Roth and Xing (1997): an applicant submits an *unordered* set of up to ten wishes; each program ranks its applicants; offers then descend those rankings over a compressed window of a few weeks. In the response phase, the applicant receives a decision on every wish—accepted, rejected, or waitlisted with a visible rank in the queue—and chooses among the offers in hand while retaining her place on the other waitlists, free to switch when a preferred offer arrives.

Two patterns in the data motivate the analysis, one for each force. First, realized choices can differ from what application portfolios would suggest. Even among students whose dominant application field matches the dominant field of their offers, 10% enroll in a program outside that field; and around 15% of students who receive an offer ultimately confirm none. Both patterns are consistent with preferences, outside options, or information changing between application and enrollment. Second, the response process displays the congestion characteristic of sequential offers: in 2019, 59% of candidates held an offer on the first day but only 28% had accepted one, after which the market cleared through a long, slow tail, with later movement concentrated among candidates lower in programs' queues. Neither fact is decisive on its own, but together they show why the welfare comparison must weigh preference flexibility against the time cost of sequential clearing.

The analysis proceeds in two parts. First, I recover the distribution of applicant valuations from the application portfolios the platform elicits. Because applicants do not rank their wishes, the data reveal which programs were chosen but not the order in which they are preferred. I model application as a portfolio problem (Chade and Smith, 2006; Ali and Shorrer, 2025): each applicant assembles the set that maximizes expected utility given her valuations for individual programs and her beliefs about admission, trading the option value of a riskier, more-preferred program against the insurance value of a safer one. In the second part, conditional on the offers a student receives, I recover her valuations at the enrollment stage.

Second, I use the recovered valuations to separate the two forces. Because congestion reflects a limited clearing window, I vary that constraint: relaxing it recovers what a frictionless sequential mechanism would deliver, while the observed constraint reproduces the congestion applicants actually face. Comparing both to the static ROL mechanism decomposes the welfare difference into a *preference learning value*—the gap between the frictionless sequential outcome and the static one—and a *congestion cost*—the gap between the frictionless and the observed sequential outcomes. The same exercise identifies which components fall on which applicants.

This tradeoff is not only of academic interest but also central to an ongoing policy debate. Beginning with the 2022 admission cycle, Parcoursup reintroduced a ranking of wishes at the later offer stage (early July), a change whose stated purpose is to accelerate clearing. This ranking has progressively been brought forward toward the earlier offer stage: in 2025, applicants are required to rank the options on their waitlist one week after the first round of offers. The same remedy was proposed at the platform's inception by Terrier et al. (2018), who identified the procedure's slowness as its principal defect and recommended that a rank-ordered list be used to assign students in a single pass. The analysis here takes that diagnosis seriously but reframes it: the slow window is also the interval during which students learn which programs are likely to admit them and gather information about those programs. Whether a single-pass ROL improves welfare therefore depends on how much preference learning it forecloses relative to the congestion it removes.

Related literature. The paper connects several strands of the school choice and matching literatures. The first is the empirical welfare evaluation of assignment mechanisms (Abdulkadiroğlu et al., 2017; Calsamiglia et al., 2020; Terrier et al., 2021; Avery et al., 2025), including dynamic designs in which offers and responses unfold over time (Klijn et al., 2019; Bó and Hakimov, 2020; Grenet et al., 2022; Gong and Liang, 2025; De Groote et al., 2025; Reischmann et al., 2021). I compare the welfare of a direct ROL mechanism to that of a sequential mechanism, adopting the counterfactual logic of Abdulkadiroğlu et al. (2017)—varying processing time—to quantify the congestion cost of the sequential design.

The paper also relates to the study of information frictions in school choice: heterogeneous beliefs about admission chances (Kapor et al., 2020; Luflade, 2018) and sequential preference learning (Grenet et al., 2022; Narita, 2018; **?**). Closest to this paper, **?** show, theoretically and in the laboratory, that a sequential mechanism that reveals an applicant's budget set raises welfare by directing costly preference learning toward attainable options. Narita (2018) documents preference change—evidenced by choice reversals—in the New York City school match and proposes a dynamic mechanism as an improvement. Both papers isolate the flexibility benefit of a sequential design; neither incorporates the congestion cost that a bounded clearing window imposes. This paper measures the two forces jointly.

The third strand is the identification of preferences from matching data. This literature adopts a range of behavioral assumptions—from truthful reporting (Abdulkadiroğlu et al., 2017), to portfolio optimality under strategic incentives (Agarwal and Somaini, 2018; Calsamiglia et al., 2020), to undominated strategies (He, 2015; Hwang, 2016; Fack et al., 2019). In a different spirit, stability-based approaches (Fack et al., 2019; He et al., 2024) are widely valued for their usefulness in applications. However, because I draw a crucial distinction between preferences at different stages, stability-based approaches are not available to me. I therefore rely on portfolio optimality; and because portfolio problems are generally difficult to handle, I simplify the framework by exploiting *riskiness* comparisons between options.

The paper proceeds as follows. Section 2 describes the setting and data. Section 3 documents the descriptive footprints of preference learning and congestion. Section 4 presents the portfolio model. Section 5 describes the estimation and preliminary results. Section 6 discusses how to use counterfactual simulation to decompose welfare consequences. Section 7 concludes.

2 Context

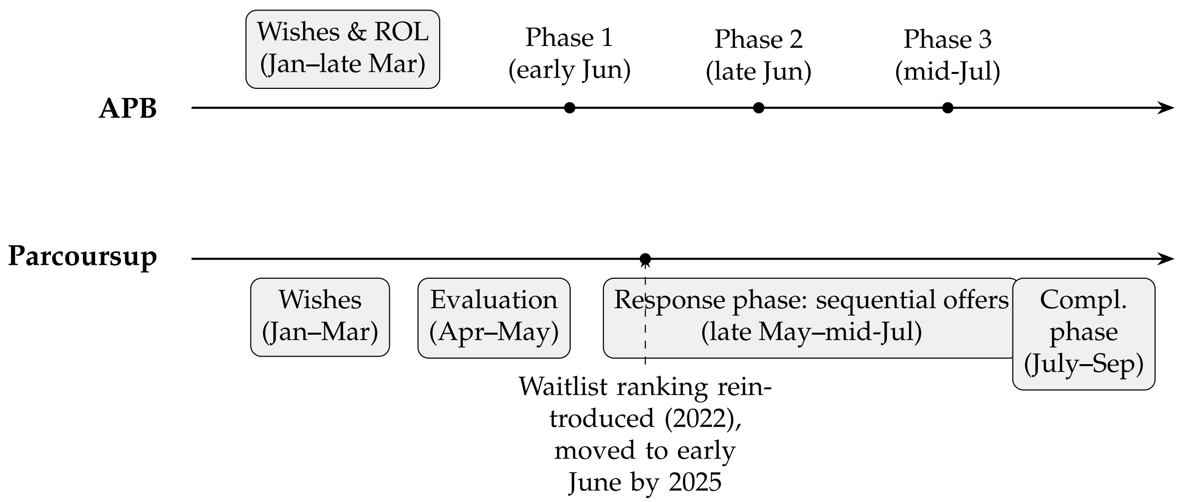
2.1 Institutional Setting

Each year, roughly 600,000 high-school graduates apply to first-year post-secondary programs in France through a centralized national platform. The menu of programs is large and heterogeneous: students choose among roughly 10,000 programs housed in more than 3,000 institutions. Over time, the platform has progressively absorbed programs that previously recruited off-platform.

Before the introduction of the platform *Parcoursup* in 2018, a system called *Admission Post-Bac* (APB) operated from 2009 to 2017. Students ranked up to 24 programs in strict order of preference, and a variant of college-proposing deferred acceptance then computed an assignment in three discrete phases (early June, late June, and mid-July), returning at most one offer per phase. All wishes ranked below an accepted offer were automatically eliminated. The mechanism was informationally opaque. First, applicants did not know how programs evaluated them, or whether a program's rank in their ROL affected their priority at that program. Second, no score or past admission cutoff was ever disclosed, nor was an applicant's own standing relative to the rest of the pool. Students therefore had little basis on which to form expectations about their admission chances.

For oversubscribed non-selective programs (primarily university *licences*), APB resolved excess demand through a weighted lottery that gave priority to applicants from the same *academie* (regional school district) and, in some cases, to those who ranked the program higher on their wish list. The lottery became politically untenable after high-profile cases in which qualified applicants were denied places in fields on the basis of a random draw. This controversy was the proximate catalyst for the replacement of APB by *Parcoursup*. The new platform introduced three fundamental changes: **unranked wishes**, **program-level selection**, and **sequential offers with visible waitlist positions**. Figure 1 illustrates the timeline of the two systems. I detail the *Parcoursup* process on which the model is based below.

Figure 1: Calendars of the two systems.



Notes: APB cleared in three discrete phases; Parcoursup runs a continuous response phase with sequential offers.

Application phase (January–March). Students submit up to ten *unranked* wishes, each accompanied by a *Fiche Avenir* (an academic report compiled by the high school) and, for some programs, a personal statement. Unlike under APB, students are not asked to rank their wishes. Sub-wishes are permitted within certain multi-campus program families, so the effective number of options listed can exceed ten.

Evaluation phase (April–May). Each program independently evaluates and ranks all applicants who listed it. For selective programs (CPGE, BTS, BUT, engineering, business), this has always been standard; the practice now extends to university *licence* programs, which rank candidates through examination committees on the basis of academic records and subject-specific criteria. These local ranking rules are not published, though the government maintains that applicants were informed of the expected competencies and general criteria. At first glance the system resembles the Common App in the United States. What differentiates it is the coordination of offers: a program's role is limited to ranking applicants, while the calling of offers is administered entirely by the central clearinghouse.

Main phase (late May–mid July). Beginning in late May, the platform returns a decision on every wish: accepted (*oui*); conditionally accepted (*oui si*—the student is admitted but asked to follow a remedial track); rejected (*non*, for selective programs only, since non-selective *licence* programs cannot reject applicants outright but may waitlist them); or waitlisted (*en attente*). Crucially, for each waitlisted wish the student observes her rank in the queue—her position among all applicants not yet offered a seat. A student holding multiple offers must accept at most one and release the rest, but she may hold her place on any number of waitlists simultaneously. Students have six days—later reduced to two—to respond to each new offer. Withdrawals cascade: when a student declines an offer, the platform extends one to the next-ranked applicant on that program's waiting list. The main phase runs continuously from late May through mid-July.

Complementary and administrative phase (late June–early September). Near the end of the main phase, a complementary phase begins, mirroring the aftermarket clearing in Roth and Xing (1997): students still seeking admission actively search for available seats, and programs have a few days to respond to each new applicant. It absorbs students who did not accept an offer during the main phase, either because they received no acceptable offers or because the offers they received are no longer desired.

Figure 8 puts together all the steps described above in a flowchart.

From 2022 onward. Recognizing the slowness of the system in its early years, Parcoursup reintroduced a ranking of waitlisted wishes in the main phase. The requirement was initially placed late in the main phase (early July 2022) and has since been moved forward to the first round of offers (early June 2025)—likely because ranking waitlisted wishes yields little benefit once most students have already confirmed an acceptance late in the phase.

2.2 Data

I use administrative records covering five cohorts of applicants from 2016 to 2020—two under *APB* (2016–2017) and three under *Parcoursup*. The summary statistics are based on the full cohorts, while the estimation uses only data from *Parcoursup*.

Students. The student file contains one record per applicant, reporting gender, scholarship status, nationality, baccalaureat track and honors, residence *academie* (regional school district), and high school of origin. I restrict the analysis sample to (i) first-time baccalaureat holders, (ii) applicants residing in Metropolitan France, and (iii) applicants who submit at least one application in the main admission phase. After these restrictions, the sample contains roughly 550,000 to 600,000 students per cohort. Its composition is stable across cohorts along most dimensions; around 20% are scholarship holders, with a 6:2:2 split across baccalaureat tracks (general, technological, and professional). Full descriptive statistics are reported in Table 6.

Programs. The program file contains one record per program, reporting the host institution, field of study, program type, public/private status, capacity, and geographic location. About three-quarters of programs belong to public institutions. There is a significant share of vocational programs (*BTS*), but they are small in size. Bachelor programs (*Licence*) account for the largest number of seats across all program types. Table 7 reports the full breakdown by type.

Wishes and offers. The Parcoursup file contains one record per (student, program) pair in each applicant's wish set. For each pair, it records: (i) the program's ranking of the applicant—the position assigned by its examination committee, together with the total number of applicants ranked, so that the relative position in the queue is observable; (ii) the sequence of decisions returned by the platform—accepted (*oui*), rejected (*non*), or waitlisted (*en attente*)—with timestamps; (iii) the applicant's responses (accept, decline, hold), also timestamped; and (iv) the final assignment indicator. The **waitlist rank** at any point can be reconstructed from the program's ranking and the offers already made. This structure is also what motivates the preference-learning hypothesis: students observe not only the decision of each program but also their rank in its queue, which allows them to direct their searches. Hakimov et al. (2023) confirm in the laboratory that revealing budget-set information in this way improves matching outcomes.

3 Descriptive Evidence

I document two patterns that motivate the analysis, using the **2019** cohort as an illustration. Neither is conclusive on its own—both require structural analysis to disentangle the mechanisms at work—but together they establish that the preference flexibility–congestion tradeoff is present in the data.

3.1 Potential preference change

Accepted field relative to predominantly applied field. Under the APB rank-ordered-list mechanism, the SIES statistical office cross-tabulates each applicant's first-ranked field

Table 1: Accepted field by dominant applied field, conditional on availability, 2019

	Final chosen field (%)							
Dominant applied field	Bachelor	Prep	BTS (Vocational)	BUT (Technical)	Engineering	Business	Care	Others
Bachelor	**89.0**	3.6	1.7	3.2	0.7	0.2	0.4	1.3
Prep	10.0	**80.0**	0.2	2.3	6.7	0.1	0.0	0.6
BTS (Vocational)	6.0	0.7	**81.8**	9.0	0.3	0.2	0.5	1.4
BUT (Technical)	9.4	3.9	2.3	**78.0**	4.4	0.4	0.2	1.3
Engineering	4.0	14.7	0.1	3.5	**77.0**	0.1	0.0	0.6
Business	3.4	0.9	0.2	1.1	0.2	**93.8**	0.0	0.4
Care	8.1	0.3	1.4	0.8	0.1	0.0	**89.1**	0.2
Others	3.6	1.0	1.9	1.4	0.2	0.1	0.1	**91.7**

Notes: Each row sums to 100%. Sample restricted to candidates who received at least one offer in their dominant applied field. Source: *Parcoursup* 2019.

against her final assignment, using the first rank as a proxy for preferences (see SIES 2016 report). Because Parcoursup does not elicit rankings, I use the dominant applied field—the field to which the candidate submitted the most wishes—as an approximation instead.

I **condition on candidates whose main application field matches** is available at the time of decisso as to avoid the impact of program admissions: because the offers available to them are aligned with their applications, any field switch at acceptance reflects the candidate's *active* decision rather than *passive* admission. Table 1 cross-tabulates the finding.

A sizable share of candidates do not choose their dominant applied field even when it is available. The diagonal entries are high but below one across all fields. Among candidates who predominantly applied to CPGE and received a CPGE offer, 88.1% accept one; 13.3% switch to engineering instead, the largest off-diagonal flow.

Search beyond the main phase. A second pattern: among candidates who receive a positive main-phase response, about 15% either withdraw from the platform or enter the complementary phase—an aftermarket for students seeking options beyond their main-phase offers. Table 2 decomposes this group. About 1.8% of candidates received a main-phase offer and yet obtained a complementary-phase offer as well (3.2% including apprenticeship), indicating active search despite an available option. A further 9.4% withdrew while still holding a positive offer. The combined share rises from 11% among general-track candidates to 17% among professional-track candidates.

Both patterns—field switching when the dominant field was available, and continued search despite a positive offer—are consistent with preferences that are not settled at application. They are the footprint of preference learning. Neither is conclusive. Separating preference learning from these alternatives is the task of the structural model.

Table 2: Candidate outcomes by proposal phase, 2019 (% of baccalaureat holders)

	Ensemble	General	Techno.	Prof.
Proposal in main phase only	77.4	85.1	72.9	55.9
Proposal in both main & compl.	**1.8**	**1.6**	**2.4**	**1.6**
incl. apprenticeship	3.2	2.2	4.2	5.3
Withdrew holding a positive offer	**9.4**	**8.9**	**8.6**	**12.1**

Notes: Rows partition all baccalaureat holders; columns sum to 100%. "Proposal in both" identifies candidates who received a main-phase offer and obtained one in the complementary phase. Source: *Parcoursup* 2019, SIES.

3.2 Congestion and offer processing

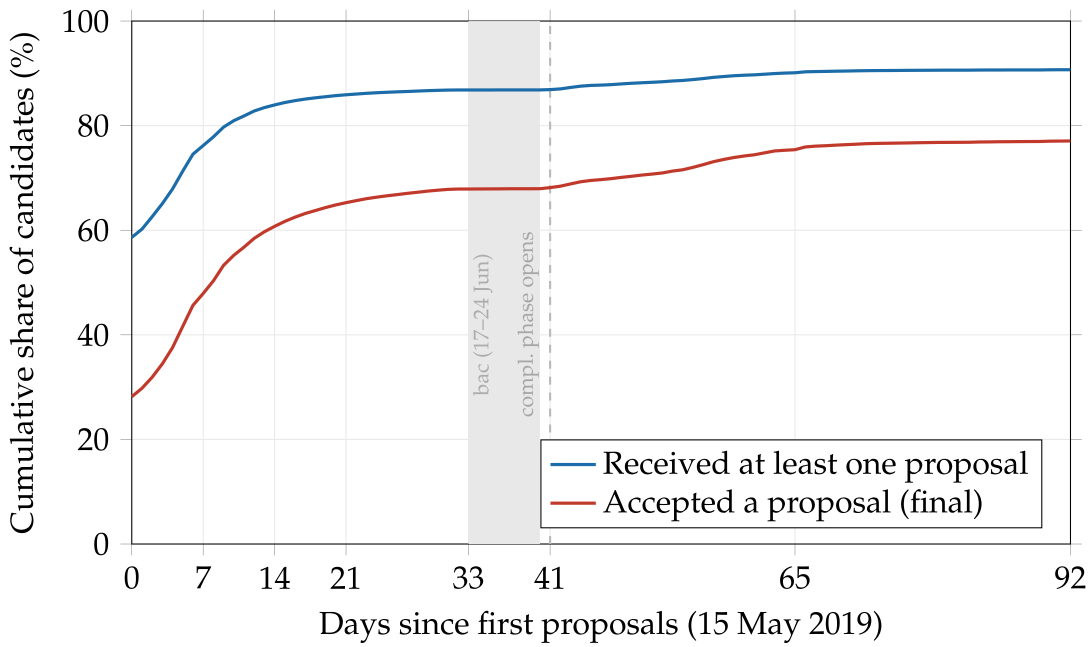
Figure 2 plots the cumulative share of candidates who have received at least one proposal and who have accepted one, by day from the opening of results (15 May 2019). The two series share a distinctive shape: rapid early growth followed by a long, slow tail. On the first day, 59% of candidates hold a proposal and 28% have accepted one; after one week, 75% and 46%; after two weeks, 84% and 60%. The pace then drops sharply. By 16 June—the day before the baccalaureat suspension—the accepted share has reached 68% but advances at barely one percentage point per week. After the baccalaureat results in early July the pace picks up briefly, reaching 75% by the close of the main phase on 19 July; the complementary phase and a further two months of processing bring the final figure to 80% by mid-September, of which 76.6 percentage points were absorbed by the main phase.

This pattern is the signature of a market that clears through sequential offers. Roth and Xing (1997) document and simulate the same shape in the decentralised market for clinical psychologists: rapid initial clearing, then a bottleneck as each further proposal requires a higher-ranked candidate to release a seat. The mechanism here is the same: on the first day every program extends an offer to its top-ranked candidate; those candidates hold multiple offers and release all but one only after the response window closes; each release triggers at most one new offer, which requires another response window before the next advance.

To confirm that the bottleneck is structural rather than an artefact of the particular calendar, I simulate a college-proposing DA using the observed applications and program rankings but with randomly drawn student ROLs over their wish sets. Table 3 reports the result. The market needs well over 200 rounds to approach convergence, with the bulk of new matches concentrated in the first rounds and a long tail thereafter—the same fast-then-slow pattern as the observed data, produced purely by the sequential clearing logic.

This time constraint has a welfare consequence. Under college-proposing DA run to

Figure 2: Cumulative share of candidates having received and having accepted a proposal, 2019



Notes: Day 0 is 15 May 2019, the first day of proposals. The shaded band marks the suspension of response deadlines during the written baccalaureat (17–24 June); the dashed line marks the opening of the complementary phase (25 June). "Accepted" counts candidates holding a definitive acceptance at end of each day. Source: *Parcoursup* 2019, SIES.

Table 3: Simulated college-proposing DA: offers and matches by round

Round	Offers sent	First offer (%)	Final chosen offer (%)
1	693,480	43.4	17.0
2	418,153	57.6	27.3
3	323,545	66.0	35.5
4	263,133	71.7	42.3
5	218,986	75.9	48.0
279	2	93.1	93.1
280	2	93.1	93.1
281	2	93.1	93.1
282	2	93.1	93.1
283	1	93.1	93.1

Notes: Simulation uses the observed application sets and program rankings of applicants. Student ROLs are drawn uniformly at random over each student's wish set. Simulation terminates at >200 rounds. Source: author's calculations.

completion, every assigned student obtains her best match among all stable matchings consistent with these program rankings (Gale and Shapley, 1962). Terminating the algorithm early leaves students low in the queue without the offer they would have received under sufficient processing time.

Table 4 confirms that the cost falls unevenly. General-track candidates received on average 4.8 proposals and waited 2.9 days for their first; 73% held a proposal on day one. Professional-track candidates received 2.5 proposals, waited 8.4 days, and only 49% had a proposal on day one. The slow tail in Figure 2 is populated disproportionately by technological and professional candidates, whose academic records place them lower in most programs' queues. The incidence is amplified by the correlation structure of program preferences: because programs weight baccalaureat honours and track similarly, the same students tend to rank low across many queues simultaneously—a feature Roth and Xing (1997) identify in simulation that can worsen bottlenecks in sequential markets.

Table 4: Proposals and waiting times by baccalaureat track, 2019

Track	Mean no. proposals	Mean days to first proposal	Proposal on day 1 (%)	Accepted a proposal (%)
General	4.77	2.88	73.2	87.5
Technological	3.42	7.06	52.7	81.8
Professional	2.52	8.40	48.9	72.3
All	4.18	4.53	65.5	84.2

Notes: Means are conditional on receiving at least one proposal. Source: *Parcoursup* 2019, SIES.

4 A model of application in the sequential mechanism

I model a student's decision on the *Parcoursup* platform as a two-stage problem: an *application stage*, in which she selects a portfolio of programs under admission uncertainty, and a *response stage*, in which offers are realized over time and she chooses among them sequentially, with the possibility of preference learning.

Timing. Let $\mathcal{J} = \{1, \ldots, J\}$ index the programs available on the platform, with $j = 0$ denoting the outside option, which captures enrollment outside the platform, a gap year, or exit from the education system; let $\bar{K}$ be the platform cap on the number of wishes a student may submit. The sequence of events is as follows.

- **Stage 1 (Student application).** Student i observes her *uninformed* utilities which is determined by her characteristics Z_i, program attributes X_j, and her private taste vector $\eta_i = (\eta_{i1}, \ldots, \eta_{iJ})$. She forms subjective beliefs about admission probabilities $\pi_i = (\pi_{i1}, \ldots, \pi_{iJ})$ and submits a portfolio $P_i \subseteq \mathcal{J}$ with $|P_i| \leq \bar{K}$ based on her uninformed preferences and beliefs.

- **Interim (School decision).** Each program j scores its non-rejected applicants on the basis of their academic records and transmits the resulting ranking to the clearing-house.

- **Stage 2 (Student response).** Offers are released over T rounds; let $O_{ijt} \in \{0, 1\}$ indicate whether i holds an active offer from j at round t. At each round, she observes the offers as well as her rank position in the waitlist. She learns a second taste vector $\varepsilon_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iJ})$ for the option she has searched. At each round she retains at most one offer while remaining on the waiting lists of the others; the procedure terminates when she accepts an offer permanently or when round T is reached. The number of response rounds, $T = \lfloor \Delta/\tau \rfloor$—where Δ is the length of the response window and τ the length of one response period—determines how many offers can be processed. It governs the degree of congestion, as in Roth and Xing (1997).

Preferences. The indirect utility student i derives from enrolling in program j, relative to the outside option normalized to $V_{i0} = 0$, is

$$V_{ij} = u(Z_i, X_j; \beta) + \Phi_{ij}, \tag{1}$$

where $u(\cdot\,; \beta)$ is a known function of student and program observables—field of study, demographics, and geography—with preference parameter vector β, and Φ_{ij} is an unobservable taste component additively separable from the systematic part. The central assumption is that this component decomposes as

$$\Phi_{ij} = \eta_{ij} + \varepsilon_{ij}, \tag{2}$$

into a part η_{ij} known to the student when she composes her portfolio—reflecting private assessments of curriculum fit, family ties to the city, or information from a campus visit—and a residual ε_{ij} learned only during the response window, once the program has become a concrete and attainable option and the student has acquired the relevant information. This decomposition formalizes the preference-learning channel that distinguishes *Parcoursup* from a direct DA mechanism: under the latter, ε_{ij} never affects the decision, whereas under *Parcoursup* the student observes ε_i before committing to enrollment. I refer to $v_{ij} = u(Z_i, X_j; \beta) + \eta_{ij}$ as the student's uninformed utility—the part of utility known at the application stage.

Selectivity. Each program j submits a strict ranking $\succ_j$ over its non-rejected applicants. Student i is admitted to j if her rank r_{ij} is at least the program's realized admission cutoff $\bar{r}_j$. Because this cutoff depends on the acceptance behavior of higher-ranked students—and because her own rank is itself uncertain ex ante—admission is not known to the applicant when she applies. From her perspective, the relevant reduced-form object is the subjective admission probability

$$\pi_{ij} = \Pr\big(r_{ij} \geq \bar{r}_j\big)\,,$$

which she forms from the publicly available statistics on each program's *Parcoursup* information sheet—the number of applicants, the number of candidates who could receive an offer, the number who ultimately enrolled, and the capacity—together with her own characteristics.[1]

I emulate the information available on these sheets. For each program j, I compute the empirical admission rate among applicants who share student i's observable profile, defined by the interaction of baccalaureat track, baccalaureat honors (a proxy for high-school grades), terminal year specialty course, and an indicator for whether the applicant resides in the same *academie* as the program. Letting $c(i)$ denote this cell, the subjective admission probability is set to

$$\hat{\pi}_{ij} \;=\; \frac{\#\{\, i' : c(i') = c(i),\ i' \text{ applied to } j,\ i' \text{ admitted to } j \,\}}{\#\{\, i' : c(i') = c(i),\ i' \text{ applied to } j \,\}},$$

the share of admitted students among applicants to j in cell $c(i)$, computed from prior-year data. Program j is more selective for student i than program k if $\pi_{ij} < \pi_{ik}$.

Rather than treating them as equilibrium objects, I take the subjective admission probabilities $\{\pi_{ij}\}$ as given, interpreting them as prices in a large market. They are subjective inputs and need not coincide with the true admission chances; operationally, I set each student's beliefs equal to the probabilities implied by these historical admission statistics.

Stage 1: Application stage. Starting with Stage 1, the student knows her uninformed utilities $(v_{ij})_{j\in\mathcal{J}}$ and her subjective admission probabilities $(\pi_{ij})_{j\in\mathcal{J}}$, but does not yet ob-

[1] See, e.g., this sample program sheet; Appendix Figures 5, 6, and 7 provide screenshots. Each program sheet reports the prior-year figures listed above and additionally allows filtering by baccalaureat track.

serve the response-stage shocks $(\varepsilon_{ij})_{j\in\mathcal{J}}$.[2]

Let $O \subseteq P$ denote the (random) set of programs from which the student holds an offer, and let $G(\cdot \mid P)$ be the distribution over offer sets O induced by the admission probabilities $(\pi_{ij})_{j\in P}$. Evaluated at her uninformed utilities, the value of an offer set is that of the best option in hand, $\max_{j\in O\cup\{0\}} v_{ij}$, so the portfolio value is

$$V(P) = \int \max_{j\in O\cup\{0\}} v_{ij}\, dG(O \mid P). \tag{3}$$

The student solves $P^\star = \arg\max_{P:\,|P|\le \bar{K}} V(P)$, trading off the utility of more-preferred programs against the admission probability of safer ones.

The portfolio-choice literature provides methods to solve this problem given its primitives (v, G), using either a greedy algorithm (Chade and Smith, 2006) or dynamic programming (Ali and Shorrer, 2025). Because no closed-form likelihood function exists for observed portfolio choices, estimation typically requires simulated likelihood, which in turn requires solving the portfolio program repeatedly at each candidate parameter value. The computational burden is substantial, particularly when the number of programs is large. I take a different approach: I rely on necessary optimality conditions of observed choices, which avoids solving the portfolio problem repeatedly in estimation.

Proposition 4.1 (Revealed preference under risk)**.** Let $P^\star$ be an optimal portfolio. Suppose $j \in P^\star$ and there exists an alternative $k \notin P^\star$ with $\pi_{ik} \ge \pi_{ij}$. Then $v_{ij} \ge v_{ik}$.

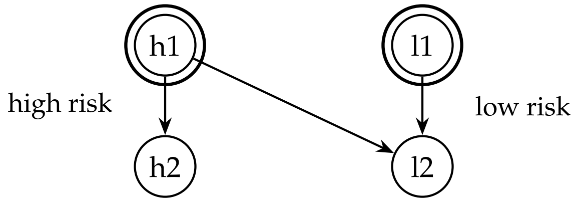
The intuition is that including a riskier program j in the portfolio while omitting a weakly safer program k can be rationalized only if the student values j at least as much as k: the higher interim valuation of j must compensate for its lower probability of yielding an offer. Each (chosen, weakly-safer-unchosen) pair yields one such inequality, and with a large menu $\mathcal{J}$ and a binding capacity constraint $\bar{K}$, each student generates many pairwise comparisons, as illustrated in Figure 3.

Only the relative selectivity across programs matters for identification, not the exact level of each π_{ij}. When necessary, programs can be grouped into R selectivity classes within which admission probabilities are treated as equal.

Stage 2: Response stage. In Stage 2, the process mirrors the inner loop of a college-proposing deferred acceptance algorithm (Roth and Xing, 1997). In each round, the student provisionally holds the best offer she currently has and declines the rest; rejected

[2] This assumption is consistent with anecdotal evidence from interviews: students report reluctance to commit to a strict ranking of programs precisely because they are unsure of their eventual preferences—unless they already know exactly where they want to go.

Figure 3: Revealed preference under risk

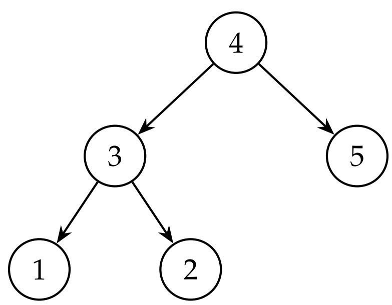


Notes: Revealed-preference comparisons from a portfolio choice with high-risk and low-risk programs. Circled options are chosen. Arrows indicate comparisons revealed by choosing an option over a weakly safer alternative; the crossed dashed line marks that the two chosen programs are not directly compared.

seats are then reallocated, and in the next round she again holds whichever option is best among the offers in hand. At this stage, there is no incentive for manipulation other than to truthfully reveal her preferences. This hold-and-decline structure is what distinguishes *Parcoursup*—a coordinated system—from the uncoordinated Common App, where there is no **provisional acceptance** and seat reallocation is not **centrally administered**.

I assume that at this stage the student's hold-and-decline decisions reveal a partial ordering of her preferences. The ordering revealed in Stage 2 is over the realized utilities $V_{ij} = v_{ij} + \varepsilon_{ij}$, since the student has by then observed ε_i. Figure 4 presents an example response path and the corresponding preference tree: two options are comparable whenever a directed path runs from one to the other; pairs with no such path remain unranked.

Figure 4: Revealed preference from a decision path



Notes: Revealed partial preference order from one response history (declined 1 and 2, held 3, switched to 4 and accepted, declined 5). An edge $a \to b$ reads "a is revealed preferred to b." Edges shown are direct comparisons; the rankings $4 \succ 1$ and $4 \succ 2$ follow by transitivity. The pairs $\{3, 5\}$ and $\{1, 2\}$ are incomparable.

4.1 Identification

The revealed-preference argument imposes minimal assumptions on behavior; in particular, it does not require students' beliefs to be cardinally accurate, only that programs be correctly ordered by relative selectivity. I first build a partial-identification argument on the revealed-preference inequalities of Proposition 4.1, in the spirit of He (2015); Hwang (2016); Fack et al. (2019).

Partial identification under an incomplete model. The portfolio setting differs from the rank-ordered-list (ROL) setting in the *source* of the comparison. In the ROL setting, the ordering is revealed *within* the submitted list—ranking program s_1 above s_2 reveals $s_1 \succ s_2$. In the portfolio setting, it is revealed *between* chosen and unchosen programs. Absent a behavioral rule that selects among optimal portfolios, the model does not pin down a unique choice for given primitives: the map from (x, u)—with x observed and u unobserved—to the outcome y is a correspondence rather than a function, so the econometric model is **incomplete** (Tamer, 2003). Proposition 4.1 delivers, for each pair (j, k) with $j \in P_i^\star$, $k \notin P_i^\star$, and $\pi_{ik} \geq \pi_{ij}$, the individual-level inequality

$$v_{ij} - v_{ik} \;=\; u(X_{ij}; \beta) - u(X_{ik}; \beta) + \eta_{ijk} \;\geq\; 0, \tag{4}$$

where $\eta_{ijk} = \eta_{ij} - \eta_{ik}$ is the unobserved taste differential. Because Proposition 4.1 characterizes a necessary optimality condition, it implies the following set inclusion

$$\left\{ v_{ij} \geq v_{ik},\ j \in P_i^\star,\ k \notin P_i^\star,\ \pi_{ik} \geq \pi_{ij} \right\} \;\subseteq\; \left\{ v_{ij} \geq v_{ik},\ \pi_{ik} \geq \pi_{ij} \right\},$$

which, after conditioning on X_i and restricting to the subpopulation for which k is weakly less selective than j, yields the lower bound

$$\Pr\left[v_{ij} \geq v_{ik} \mid X_i,\ \pi_{ik} \geq \pi_{ij}; \beta\right] \;\geq\; \Pr\left[j \in P_i^\star,\ k \notin P_i^\star \mid X_i,\ \pi_{ik} \geq \pi_{ij}\right]. \tag{5}$$

A symmetric argument delivers an upper bound. Among students for whom j is weakly less selective than k, choosing k while not j reveals $k \succ j$, so

$$\Pr\left[v_{ij} \geq v_{ik} \mid X_i,\ \pi_{ij} \geq \pi_{ik}; \beta\right] \;\leq\; 1 - \Pr\left[k \in P_i^\star,\ j \notin P_i^\star \mid X_i,\ \pi_{ij} \geq \pi_{ik}\right]. \tag{6}$$

I impose the following assumptions:

(A1) Representative discriminating subsample. Fix a pair (j, k) and condition on X_i. Among the four portfolio configurations—neither, both, only j, or only k—the subpopulation in which exactly one of j and k is chosen is representative of the population in its j-versus-k preferences; that is, restricting to it does not select on the sign of $v_{ij} - v_{ik}$. Within this subpopulation the two choice shares sum to one,

$$\Pr\left[j \in P_i^\star,\ k \notin P_i^\star \mid X_i\right] + \Pr\left[k \in P_i^\star,\ j \notin P_i^\star \mid X_i\right] = 1.$$

(A2) Selectivity exogeneity. Conditional on X_i, the preference event $\{v_{ij} \geq v_{ik}\}$ is independent of relative selectivity, so that

$$\Pr\left[v_{ij} \geq v_{ik} \mid X_i,\, \pi_{ij} \geq \pi_{ik}; \beta\right] = \Pr\left[v_{ij} \geq v_{ik} \mid X_i,\, \pi_{ik} \geq \pi_{ij}; \beta\right] = \Pr\left[v_{ij} \geq v_{ik} \mid X_i; \beta\right].$$

Using (A1) to rewrite the right-hand side of (6) as a j-chosen share, and (A2) to replace the conditional preference probabilities with $\Pr(v_{ij} \geq v_{ik} \mid X_i; \beta)$, the two bounds combine into

$$\Pr\left[j \in P_i^\star,\, k \notin P_i^\star \mid X_i,\, j \text{ riskier}\right] \;\leq\; \Pr\left[v_{ij} \geq v_{ik} \mid X_i; \beta\right] \;\leq\; \Pr\left[j \in P_i^\star,\, k \notin P_i^\star \mid X_i,\, k \text{ riskier}\right]. \tag{7}$$

Both outer terms in (7) are observed in the data, whereas the middle term is model-implied: it depends on β through the systematic-utility difference $u(X_{ij}; \beta) - u(X_{ik}; \beta)$ and on the distribution of the taste differential η_{ijk}.

Interacting the conditional inequalities (5)–(6) with a nonnegative instrument function $h(X_i)$ and integrating over X_i yields unconditional moment inequalities. From the lower bound (5),

$$\mathbb{E}\big[\big[\Pr(v_{ij} \geq v_{ik} \mid X_i; \beta) - \mathbf{1}\{j \in P_i^\star,\, k \notin P_i^\star\}\big] \cdot \mathbf{1}\{\pi_{ik} \geq \pi_{ij}\} \cdot h(X_i)\big] \;\geq\; 0. \tag{8}$$

$$\mathbb{E}\big[\big[\mathbf{1}\{j \in P_i^\star,\, k \notin P_i^\star\} - \Pr(v_{ij} \geq v_{ik} \mid X_i; \beta)\big] \cdot \mathbf{1}\{\pi_{ij} \geq \pi_{ik}\} \cdot h(X_i)\big] \;\geq\; 0. \tag{9}$$

The first term inside the bracket is model-implied and depends on β; the second is the observed choice indicator. Each (chosen, weakly-safer-unchosen) pair contributes one such inequality.

Point identification under more restrictions. The moment inequalities (8) characterize an identified set for β. To make explicit which assumptions collapse the set to a point, it is convenient to recast the problem as one of a censored binary outcome. Fix a pair (j, k) and condition throughout on X_i. Let $A_i = \mathbf{1}\{v_{ij} \geq v_{ik}\}$ denote the latent preference indicator, whose conditional probability

$$p(X_i; \beta) \;\equiv\; \Pr\left[A_i = 1 \mid X_i; \beta\right] \;=\; F_\eta\left[u(X_{ij}; \beta) - u(X_{ik}; \beta)\right]$$

is the object the structural model restricts, with F_η the distribution of $-\eta_{ijk}$. The data reveal A_i only on a selected subsample. Define the revelation variable

$$T_i = \begin{cases} +1 & \text{if } j \in P_i^\star,\ k \notin P_i^\star,\ \pi_{ik} \geq \pi_{ij}, \\ -1 & \text{if } k \in P_i^\star,\ j \notin P_i^\star,\ \pi_{ij} \geq \pi_{ik}, \\ 0 & \text{otherwise.} \end{cases} \tag{10}$$

Proposition 4.1 and its mirror image imply, with no assumptions beyond correct ordinal beliefs, that $\{T_i = +1\} \subseteq \{A_i = 1\}$ and $\{T_i = -1\} \subseteq \{A_i = 0\}$: the revealed sign is never wrong. Consequently $\{T_i = +1\} = \{A_i = 1\} \cap \{T_i \neq 0\}$, and the entire gap between the bounds (5)–(6) is a selection problem: the data identify $\Pr[A_i = 1 \mid X_i, T_i \neq 0]$, the preference share within the subpopulation whose portfolio discriminates the pair, whereas the model restricts $\Pr[A_i = 1 \mid X_i]$. **Point identification is exactly the statement that the inferable subsample is representative**,

$$A_i \perp \mathbf{1}\{T_i \neq 0\} \mid X_i, \tag{11}$$

a missing-at-random condition on the censored outcome A_i.

Condition (11) is opaque as stated, because T_i bundles three distinct sources of variation: **the portfolio decision, the relative risk, and the preference itself**. Let $E_i = \mathbf{1}\{|P_i^\star \cap \{j,k\}| = 1\}$ indicate the *discriminating configuration* that exactly one member of the pair is applied to—and $R_i = \mathbf{1}\{\pi_{ik} \geq \pi_{ij}\}$ the *risk configuration*, so that $\{T_i \neq 0\}$ is the event that the portfolio discriminates the pair *and* the omitted member is the weakly safer one.

I replace (11) by three primitive conditions.

(I1) Risk configuration exogeneity. $A_i \perp R_i \mid X_i$: conditional on observables, which member of the pair is riskier carries no information about which is preferred. If subjective admission probabilities are functions of observables—student scores and program selectivity, both included in X_i—then R_i is degenerate given X_i and (I1) is vacuous; it has content only through unobserved belief heterogeneity.

(I2) Discriminating configuration exogeneity. $A_i \perp E_i \mid X_i, R_i$: whether the portfolio discriminates the pair—as opposed to including both members or neither—is not selected on the *sign* of the taste differential. The configuration E_i may depend on the levels (v_{ij}, v_{ik}), on the rest of the choice set, and on the capacity constraint $\bar{K}$; (I2) requires only that none of these channels correlate with the **sign** of $v_{ij} - v_{ik}$ given (X_i, R_i).

(I4) Index and rank. F_η is known, continuous, and strictly increasing (its scale fixed by

the application-stage variance normalization), and the map $\beta \mapsto u(X_{ij};\beta) - u(X_{ik};\beta)$ is injective on the support of the covariate differences; in the linear case, $\mathbb{E}\left[(X_{ij} - X_{ik})(X_{ij} - X_{ik})'\right]$ has full rank.

Under (I4), $u(X_{ij};\beta) - u(X_{ik};\beta) = F_\eta^{-1}\left[\Pr(A_i = 1 \mid X_i)\right]$ is identified pointwise and β is recovered from variation in covariate differences.

5 Estimation

5.1 Specification

For student i of baccalaureat track $g(i)$ and program j, the application-stage (uninformed) and response-stage (informed) utilities are

$$v_{ij} = x_{ij}' \gamma_{g(i)}^{\text{app}} + \delta_{g(i)} d_{ij} + \eta_{ij}, \qquad \eta_{ij} \sim \mathcal{N}(0,1), \tag{12}$$

$$V_{ij} = x_{ij}' \gamma_{g(i)}^{\text{enr}} + \delta_{g(i)} d_{ij} + \eta_{ij} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{g(i)}^2), \tag{13}$$

where x_{ij} is a vector of program-type dummies (CPGE, BTS, BUT, business school, engineering school, health-and-social-care, and other), and d_{ij} is the distance between student i and program j. The shock η_{ij} enters both stages; ε_{ij} is an additional unobservable realized only during the response phase, with dispersion $\sigma_{g(i)}^2$. The application-stage variance is normalized to one. All parameters—type effects γ_g^{app}, γ_g^{enr}, the distance coefficient δ_g, and the enrollment dispersion σ_g^2—are allowed to differ across the general, technological, and professional tracks. The distance coefficient is shared across stages: the same δ_g enters both v_{ij} and V_{ij}, so only one distance parameter is identified per track. Bachelor (*Licence*) serves as the reference category; type coefficients are read as utility differences relative to a bachelor program.[3]

5.2 Estimation strategy

The revealed-preference argument of Section 4.1 delivers, for each (chosen, weakly-safer-unchosen) pair, an inequality on interim valuations, and the hold-and-decline path delivers an analogous ordering at the response-stage. Depending on the assumptions imposed, these (in)equalities can be used to construct moment conditions.

[3]The outside option is currently excluded from the estimation. The treatment of the outside option needs to be determined.

The revealed-preference argument of Section 4.1 delivers, for each (chosen, weakly-safer-unchosen) pair, an inequality on uninformed utilities, and the hold-and-decline path delivers an analogous ordering at the response stage. Depending on the assumptions imposed, these (in)equalities can be used to construct moment conditions. However, treating the two stages in isolation ignores the correlation of the unobservables between the application shock η_{ij} and the response shock Φ_{ij}. This is restrictive, because the offer set O_i is endogenous: students whose application-stage tastes are high for selective programs are more likely to apply to—and therefore to receive offers from—those programs, so the composition of O_i is correlated with η_{ij} and, through it, with response-stage utility. Conditioning on the realized offer set then induces a selection problem at the response stage (Horowitz, 1991). To account for this correlation, there are two ways: (1) introduce unobserved types for each individual; (2) estimate the two stages jointly with a simulation-based method.

For the moment, I adopt the data-augmented Gibbs sampler following Abdulkadiroğlu et al. (2017); Agarwal and Somaini (2018); He et al. (2024), an ordered-choice analog of Rossi et al. (1996). Assuming normal taste shocks allows the two stages to be estimated jointly with a freely correlated $(\eta_{ij}, \varepsilon_{ij})$. The observed behavior does not reveal the latent utilities directly; it reveals inequalities among them, which restrict each student's utility vector to a truncated region. Two kinds of comparison enter. At the application stage, Proposition 4.1 turns each (chosen, weakly-safer-unchosen) pair into the restriction $v_{ij} \geq v_{ik}$. At the response stage, the mechanical hold-and-decline path orders the offers the student receives: by transitivity along the held chain, the finally accepted offer a_i dominates every offer ever declined, $V_{ia_i} \geq V_{ik}$ for all declined k.

The sampler alternates two blocks. Given the parameters, it draws each student's latent utilities from their full conditionals—univariate truncated normals imposing the inequalities above—thereby imputing a complete utility vector consistent with her observed choices. Given the imputed utilities, the parameters $(\gamma^{\text{app}}, \gamma^{\text{enr}}, \delta, \sigma^2)$ are updated from their conjugate posteriors. Iterating and discarding a burn-in yields draws from the posterior; posterior means serve as point estimates, with the same large-sample interpretation as maximum likelihood. A by-product is exactly what the counterfactuals require: for each student, the sampler returns a draw of her full latent utility vector $(V_{ij})_j$, consistent with both her application choices and her response path. Averaging over draws gives the individual utility vector used to evaluate $W(\mu)$ in (14).

5.3 Results (Preliminary)

Table 5 reports posterior means and posterior standard deviations for all three baccalaureat tracks.

Dispersion of unobservables. The estimated σ^2 is 15.311 for the general track and substantially lower for the technological (3.837) and professional (4.337) tracks. Since the application-stage variance is normalized to one, σ^2 measures the additional dispersion in preferences realized only at the response stage. For general-track students, the enrollment shock is several times as large as the application-stage shock, indicating that a sizable share of the preferences that ultimately determine enrollment is not yet resolved at the time of application.

Distance. The distance coefficient is negative and precisely estimated across all three tracks (-0.006, -0.007, and -0.008 for the general, technological, and professional tracks, respectively). Because δ_g is shared across stages, it does not admit a separate enrollment estimate.

Whether these preference shifts are large enough to reorder cardinal valuations—and hence to alter the assignment—is an empirical question that the counterfactual decomposition in Section 6 takes up.

6 Counterfactual (Discussion)

The structural estimates yield draws of the first-stage value v_{ij} and the second-stage value V_{ij} for each student–program pair. I construct three matchings, all evaluated at the realized utility V_{ij}, and use their welfare differences to decompose the gap between the sequential and static mechanisms.

Three matchings. In each case the observed application sets and the programs' rankings of applicants are held fixed; only student preferences and the clearing rule vary. Let $\mathcal{I}$ denote the set of students.

- μ^1: *Static ROL.* Students rank by first-stage utility $v_{ij} = u(X_{ij}; \beta) + \eta_{ij}$. College-proposing DA is run to completion.
- μ^2: *Full-information benchmark.* Students rank by second-stage utility V_{ij}. College-proposing DA is run to completion. This is the outcome that both a frictionless sequential mechanism and a static mechanism with well-informed preferences would deliver.

Table 5: Posterior Estimates by Baccalaureate Track

	General		Technological		Professional	
	Application	Enrollment	Application	Enrollment	Application	Enrollment
CPGE	1.023	2.907	0.964	2.652	0.082	0.705
	(0.031)	(0.328)	(0.072)	(0.276)	(0.224)	(1.239)
BTS	−1.273	−0.312	0.675	0.416	1.467	0.228
	(0.024)	(0.303)	(0.026)	(0.135)	(0.051)	(0.320)
BUT	0.830	2.099	1.969	2.002	1.904	3.232
	(0.031)	(0.307)	(0.049)	(0.184)	(0.167)	(0.737)
Business school	1.350	3.735	0.984	3.508	−1.150	6.767
	(0.065)	(0.645)	(0.106)	(0.762)	(0.472)	(5.037)
Engineering school	2.403	3.797	1.903	3.567	0.567	1.630
	(0.041)	(0.458)	(0.105)	(0.503)	(0.812)	(2.675)
Health and social care	−0.476	1.539	1.644	2.514	2.337	5.330
	(0.040)	(0.637)	(0.043)	(0.340)	(0.072)	(1.045)
Other	−0.432	2.525	0.737	2.160	1.128	−0.380
	(0.047)	(0.510)	(0.050)	(0.323)	(0.084)	(0.513)
Distance (km)	−0.006	–	−0.007	–	−0.008	–
	(0.000)	–	(0.000)	–	(0.000)	–
σ^2	1	15.311	1	3.837	1	4.337
	–	(2.691)	–	(0.573)	–	(1.206)

Bachelor is the reference program type. Application-stage variance is normalized to 1. Posterior SD in parentheses.

Notes: Gibbs sampler estimates. CPGE denotes *classes preparatoires aux grandes ecoles*. BTS denotes *brevet de technicien superieur*. BUT denotes *bachelor universitaire de technologie*. Licence is the omitted program type. Application-stage variance normalized to 1. Posterior standard deviations in parentheses.

μ^{SQ}: *Status quo.* The observed *Parcoursup* assignment, produced under second-stage responses but subject to the actual time constraint.

Decomposition. Let $W(\mu) = |\mathcal{I}|^{-1} \sum_i V_{i,\mu(i)}$. The welfare difference between the sequential and static mechanisms decomposes as

$$\underbrace{W(\mu^{SQ}) - W(\mu^1)}_{\text{net value}} = \underbrace{W(\mu^2) - W(\mu^1)}_{\text{flexibility value}} - \underbrace{W(\mu^2) - W(\mu^{SQ})}_{\text{congestion cost}}. \tag{14}$$

The flexibility value is the gain from ranking with informed rather than uninformed preferences, holding fixed that the market clears fully. The congestion cost is the loss from operating the sequential mechanism under a binding time constraint; it is weakly positive, student by student, by the receiver-side monotonicity of college-proposing DA—each student's held offer only improves as more rounds are processed, so truncation can only lower realized utility. The net value determines whether the sequential design raises welfare relative to the static one.

6.1 A mechanical measure of congestion

The decomposition in (14) charges the gap $W(\mu^2) - W(\mu^{SQ})$ to the binding clearing window. This subsection gives direct evidence that the window binds, by measuring how many offer–response rounds college-proposing deferred acceptance needs to clear the realized market. I hold fixed the observed application sets and the programs' rankings of applicants. Because *Parcoursup* elicits unordered wish sets, students' rankings over their own programs are not observed; for this exercise I impute them, drawing for each application an independent uniform rank that stands in for the student's ordinal preference. The exercise is mechanical in the sense of Abdulkadiroğlu et al. (2017): it models no behavioral response and is meant only to expose the offer-processing dynamics, not to predict the assignment.

I then run college-proposing deferred acceptance and record activity round by round. A *round* is defined in the following way: at its start, every program with a vacant seat extends offers down its ranking until its capacity is filled or its remaining applicants are exhausted; each applicant holding more than one offer retains the program she ranks highest and declines the rest; and every declined seat reopens for the next round. The market has cleared when a round produces no new declines, at which point the held assignment is the college-optimal stable matching. The round count thus measures how many sequential offer–response cycles separate the first wave of offers from a stable outcome.

Clearing the realized market takes more than 200 rounds, with a long, thin tail in which each round places only a handful of applicants. This shape reproduces the observed response path of Section 3, where two weeks of rapid movement give way to clearing at about one percentage point per week. The bulk of the market clears quickly; the congestion lives in the tail.

The round count becomes a cost only once each round is priced in calendar time. Under a centralized static mechanism the count is immaterial—deferred acceptance is computed in a single pass and convergence is free. Under *Parcoursup* it is not, because a program cannot reissue a seat until the applicant holding it has responded, so each round consumes the time applicants take to act. Treating one round as roughly one day—the cadence at which the platform releases offers—more than 250 rounds already exceeds the few-week main phase. And one day per round is optimistic: applicants neither respond simultaneously nor immediately, taking on the order of two to six days, which lengthens the implied clearing time several-fold.

This is congestion in the sense of Roth and Xing (1997): clearing time is the number of offer–response cycles times the per-cycle turnaround, and when that product exceeds the market's window the mechanism either lengthens the calendar or settles for an incomplete match. The truncation it settles for is exactly what separates μ^{SQ} from μ^2 and gives the congestion cost in (14) its sign.

Remarks. The application set is held fixed across counterfactuals, so the decomposition isolates the ranking channel from the portfolio channel: it measures the value of flexibility given the applications students made, not the value of re-optimizing the portfolio under a different mechanism. Finally, the decomposition is additive; whether the students who benefit from flexibility are those who bear the congestion cost is an empirical question I should address by reporting both terms separately by student characteristics.

7 Conclusion

This paper studies the tradeoff between flexibility for preference learning and congestion in centralized matching. A sequential mechanism lets applicants respond to information as offers arrive rather than committing to a ranking ex ante, but clearing in real time consumes calendar time and can leave lower-ranked applicants without the offer they would have received under a static ROL. *Parcoursup*, which assigns essentially all of France's annual high school graduates through sequential offers with visible waitlist positions, provides an unusual opportunity to study this tradeoff: the platform records not only applications and assignments but the full response path—every offer, waitlist rank, and decision, with timestamps—so that both the scope for preference learning and the pace of clearing are directly observable.

The descriptive evidence shows that both forces are present. On the learning side, realized choices depart from what application portfolios would suggest: even among students whose dominant application field matches the dominant field of their offers, about one in ten enrolls outside that field, and around 15% of students who receive an offer confirm none. On the congestion side, the response phase displays the signature of sequential clearing documented by Roth and Xing (1997): 59% of candidates held an offer on the first day of the 2019 campaign but only 28% had accepted one, and after two weeks of rapid movement the market advanced through a long, slow tail at roughly one percentage point per week. The tail is populated disproportionately by technological- and professional-track candidates, who wait nearly three times as long as general-track candidates for a first offer—so whatever the aggregate net value of the sequential design, its congestion cost is not borne evenly.

To weigh the two forces, I recover latent utilities at both stages. Because *Parcoursup* elicits unordered wish sets, the application stage is identified from a revealed-preference argument: choosing a riskier program over a weakly safer unchosen one reveals a higher interim valuation. The response stage is identified from the hold-and-decline path itself. The preliminary estimates indicate that the two stages order program types broadly similarly while differing in scale: under the normalization that the disutility of distance is common across stages, the response-stage taste shock is roughly three to four times as dispersed as the application-stage shock. Taken at face value, this gap says that a substantial share of the preferences that ultimately determine enrollment is learned after applications are submitted—the information the sequential design exists to accommodate.

These results carry several caveats. The two stages are currently estimated separately, imposing independence between the application-stage and response-stage unobservables; because the offer set is endogenous to application-stage tastes, this restriction induces a selection problem at the response stage. The outside option is introduced through artificial comparisons rather than observed directly. And the scale comparison rests on the normalization that the distance coefficient is common across stages. The next step is joint estimation of the two stages by Gibbs sampling with data augmentation, following Abdulkadiroğlu et al. (2017), which delivers, for each student, a latent utility vector consistent with both her application set and her response path. With these in hand, the three-matching decomposition of Section 6 quantifies the flexibility value and the congestion cost, their net, and their incidence across student groups.

The policy stakes are immediate. Since 2022, *Parcoursup* has progressively reintroduced a ranking of waitlisted wishes—moved to the first round of offers in 2025—with the stated purpose of accelerating clearing, the remedy proposed at the platform's inception by Terrier et al. (2018). Whether this is an improvement depends on the same quantity this paper measures: a single-pass or accelerated mechanism removes congestion but forecloses the preference learning that the response window permits. If the flexibility value is small, the reform is close to a free lunch; if it is large, the reform trades a visible cost (slow clearing) for a hidden one (commitments made on less informed preferences), borne by whoever would have learned the most. More broadly, the flexibility–congestion tradeoff arises in any matching market that clears through offers in real time, and the French setting, with its complete record of the response process, offers a rare opportunity to put numbers on it.

References

Abdulkadiroğlu, A., Agarwal, N., and Pathak, P. A. (2017). The welfare effects of coordinated assignment: Evidence from the new york city high school match. *American Economic Review*, 107(12):3635–3689.

Agarwal, N. and Somaini, P. (2018). Demand analysis using strategic reports: An application to a school choice mechanism. *Econometrica*, 86(2):391–444.

Ali, S. N. and Shorrer, R. I. (2025). Hedging when applying: Simultaneous search with correlation. *American Economic Review*, 115(2):571–598.

Avery, C., Kocks, G., and Pathak, P. A. (2025). The algorithm advantage: Ranked application systems outperform decentralized and common applications in boston and beyond. Technical report, National Bureau of Economic Research.

Bó, I. and Hakimov, R. (2020). Iterative versus standard deferred acceptance: Experimental evidence. *The Economic Journal*, 130(626):356–392.

Calsamiglia, C., Fu, C., and Güell, M. (2020). Structural estimation of a model of school choices: The boston mechanism versus its alternatives. *Journal of Political Economy*, 128(2):642–680.

Chade, H. and Smith, L. (2006). Simultaneous search. *Econometrica*, 74(5):1293–1307.

De Groote, O., Fabre, A., Luflade, M., and Maurel, A. (2025). Sequential college admission mechanisms and off-platform options. Technical report, National Bureau of Economic Research.

Fack, G., Grenet, J., and He, Y. (2019). Beyond truth-telling: Preference estimation with centralized school choice and college admissions. *American Economic Review*, 109(4):1486–1529.

Gale, D. and Shapley, L. S. (1962). College admissions and the stability of marriage. *The American mathematical monthly*, 69(1):9–15.

Gong, B. and Liang, Y. (2025). A dynamic matching mechanism for college admissions: Theory and experiment. *Management Science*, 71(5):4396–4412.

Grenet, J., He, Y., and Kübler, D. (2022). Preference discovery in university admissions: The case for dynamic multioffer mechanisms. *Journal of Political Economy*, 130(6):1427–1476.

Hakimov, R., Kübler, D., and Pan, S. (2023). Costly information acquisition in centralized matching markets. *Quantitative Economics*, 14(4):1447–1490.

He, Y. (2015). Gaming the boston school choice mechanism in beijing.

He, Y., Sinha, S., and Sun, X. (2024). Identification and estimation in many-to-one two-sided matching without transfers. *Econometrica*, 92(3):749–774.

Horowitz, J. L. (1991). Modeling the choice of choice set in discrete-choice random-utility models. *Environment and Planning a*, 23(9):1237–1246.

Hwang, S. (2016). A robust redesign of high school match. *EAI Endorsed Trans. Serious Games*, 3(11):e5.

Kapor, A. J., Neilson, C. A., and Zimmerman, S. D. (2020). Heterogeneous beliefs and school choice mechanisms. *American Economic Review*, 110(5):1274–1315.

Klijn, F., Pais, J., and Vorsatz, M. (2019). Static versus dynamic deferred acceptance in school choice: Theory and experiment. *Games and Economic Behavior*, 113:147–163.

Luflade, M. (2018). The value of information in centralized school choice systems. *Duke University*, 4(6):7.

Narita, Y. (2018). Match or mismatch? learning and inertia in school choice. *Learning and Inertia in School Choice (June 18, 2018)*.

Reischmann, T., Klein, T., and Giegerich, S. (2021). An iterative deferred acceptance mechanism for decentralized, fast and fair childcare assignment. Technical report, ZEW Discussion Papers.

Rossi, P. E., McCulloch, R. E., and Allenby, G. M. (1996). The value of purchase history data in target marketing. *Marketing Science*, 15(4):321–340.

Roth, A. E. and Xing, X. (1997). Turnaround time and bottlenecks in market clearing: Decentralized matching in the market for clinical psychologists. *Journal of political Economy*, 105(2):284–329.

Tamer, E. (2003). Incomplete simultaneous discrete response model with multiple equilibria. *The Review of Economic Studies*, 70(1):147–165.

Terrier, C., Combe, J., Hiller, V., and Tercieux, O. (2018). Faut-il sauver les algorithmes d'affectation ? Note, Terra Nova, Paris. Published 6 June 2018.

Terrier, C., Pathak, P. A., and Ren, K. (2021). From immediate acceptance to deferred acceptance: effects on school admissions and achievement in england. Technical report, National Bureau of Economic Research.

A Additional Tables and Figures

Table 6: Student Sample Summary Statistics

	APB		Parcoursup		
	2016	2017	2018	2019	2020
Number of students	543,544	561,954	589,815	586,849	638,962
Composition (%)					
Scholarship recipient	17.8	18.2	19.4	20.7	23.9
Baccalaureat track (%)					
General	57.7	58.3	57.9	57.6	57.0
Professional	21.4	21.0	21.7	21.8	21.7
Technical	20.9	20.7	20.4	20.6	21.4
Baccalaureat honors (%)					
Highest honors	9.1	9.6	9.2	8.6	12.0
High honors	16.8	16.7	16.8	16.7	22.2
Honors	30.8	29.8	29.9	29.7	33.4
Pass	43.3	44.0	44.2	45.1	32.4

Notes: Year 2020 exibits a larger share of higher honors in baccalaureat results, affected by the pandemic.

Table 7: Program Sample Summary Statistics

	APB		Parcoursup		
	2016	2017	2018	2019	2020
Number of programs	11,931	12,362	13,105	14,377	16,841
Number of institutions	3,396	3,501	3,656	4,164	4,642
Characteristics (%)					
Public	75.5	75.1	75.3	72.8	70.1
Scholarship eligible	86.2	83.5	78.8	76.2	71.3
Selective	70.6	70.6	81.7	82.7	61.8
Program type (%)					
CPGE	7.4	7.2	6.9	6.2	5.4
BTS, production	20.9	21.7	21.2	20.1	19.1
BTS, services	28.1	27.9	27.1	25.7	26.3
BUT, production	4.6	4.4	4.1	3.8	3.3
BUT, services	3.1	3.0	2.9	2.7	2.3
Business school	0.5	0.4	0.4	0.4	0.8
Engineering school	2.6	3.3	2.9	3.1	2.9
Licence: arts, letters, languages	8.2	8.0	7.5	7.4	6.8
Licence: law, economics, management	2.9	2.9	2.8	2.8	3.0
Licence: science, technology, health	5.2	4.4	4.3	3.7	5.3
Licence: social sciences	3.8	3.6	3.5	3.4	3.4
Health and social care	1.0	1.5	1.4	4.9	4.9

Notes: CPGE denotes *classes preparatoires aux grandes ecoles*, a two-year preparatory course for competitive entrance exams. BTS denotes *brevet de technicien superieur*, a two-year vocational program. BUT denotes *bachelor universitaire de technologie*, a two-year technical program (formerly called DUT, *diplôme universitaire de technologie*). Shares by program type are calculated using the count of programs, not the number of seats.

B Additional Figures

Figure 5: Example Parcoursup Program Sheet: Access Statistics

Figure 6: Example Parcoursup Program Sheet: Profile-Specific Information

Figure 7: Example Parcoursup Program Sheet: Admission Chances

Sur les 3 dernières années, le constat est le suivant :

LYCÉENS BAC GÉNÉRAL DU SECTEUR

Pour une moyenne générale de
10 / 20

RAREMENT OCCASIONNELLEMENT RÉGULIÈREMENT **PLUS DE 50%** PLUS DE 80%

La formation a envoyé des propositions d'admission à **plus de 50%** des lycéens de terminale générale avec ce profil.

Et cette doublette de spécialités

SES
Mathématiques

Conseils

Il est important de noter que cette donnée est un constat des années passées et ne garantit donc pas votre propre admission.

N'hésitez pas à postuler **si la formation vous intéresse**, mais pensez également à ajouter d'autres vœux de sécurité.

Quelle importance est donnée aux résultats scolaires par cette formation ?

Lors de l'analyse des dossiers, cette formation **accorde 60% d'importance** aux résultats scolaires des candidats.
Cela signifie que d'autres critères sont pris en compte par la formation dans l'examen des candidatures.

Consulter la liste complète des critères généraux d'examen des vœux

Figure 8: College-proposing deferred acceptance run over a bounded clearing window.